SMOKE Emissions Processing

AIRPACT January 2025 Simulation

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Completed Emission Sectors

6 sectors processed for 31 days (January 1-31, 2025)

- **Onroad RPD**
  Mobile vehicle emissions, RPD, RPV
- **Residential Wood Combustion**
  RWC heating sources
- **Area Fugitive Dust**
  Afdust from roads/surfaces, construction
- **Nonroad**
  Construction equipment
- **Rail**
  Locomotive emissions
- **Point Sources**
  Wildfire (pt_wildfire)

Output Location

```
merged_emis/2025/
emis_mole_merged_
 YYYYMMDD_AIRPACT_04km.ncf
```

Grid Specifications

Domain: AIRPACT_04km
Resolution: 4 km
Grid: 285 x 258 cells
Mechanism: CB6r5

SMOKE Processing Workflow

1. **Inventory**
   NEI 2022
   base year data
2. **Temporal**
   Hourly
   allocation
3. **Spatial**
   Gridding to
   AIRPACT 4km
4. **Speciate**
   CB6r5
   mechanism
5. **Merge**
   Combine
   all sectors

Key Tools

SMOKE v5.2.1 - Emissions processing
Mrggrid - Sector merging
Intel Compilers - HPC compilation
NetCDF/I/O API - Data formats

Input Data

NEI 2022 v1 inventory
MCIP meteorology from WRF
Temporal profiles (hourly/weekly)
Spatial surrogates for WA/OR/ID

On-road Emission Processing

MOVES-based Rate-Per-Distance mobile source emissions

Processing Pipeline

1. **MOVES Lookup Tables**
   Emission rates by vehicle type, speed, temp
2. **Smkinven**
   Read activity data (VMT by road type)
3. **Grdmat**
   Spatial allocation using road networks
4. **Temporal**
   Hourly/weekly traffic patterns
5. **Movesmrg**
   Apply rates × activity × met adjustments

Key Inputs

MOVES4 emission rate tables
VMT by county, road type, vehicle
Meteorology (temp, humidity)
Temporal profiles (diurnal/weekly)

Output Files (RPD, RPV)

Location:
`premerged/onroad/RPD/`

Filename pattern:
`emis_mole_RPD_onroad_YYYYMMDD_AIRPACT_04km_cmaq_cb6ae7_2022hc.ncf`

Residential Wood Combustion

Temperature-dependent heating emissions from wood stoves & fireplaces

Processing Pipeline

1. **Smkinven**
   Read annual RWC inventory by county
2. **Grdmat**
   Spatial allocation (population surrogate)
3. **Spcmat**
   Chemical speciation to CB6r5
4. **Temporal + Met**
   Temperature-based adjustment factors
5. **Smkmerge**
   Combine and output hourly emissions

Temperature Adjustment

Emissions scale with heating demand:

- Below 50°F: Full emissions
- 50-60°F: Scaled emissions
- Above 60°F: Minimal emissions

Key RWC Species

PM2.5 - Fine particulates
POC - Primary organic carbon
PEC - Primary elemental carbon
CO, VOCs, PAHs

Output:
`premerged/rwc/emis_mole_rwc_YYYYMMDD_*.ncf`

Wildfire Emissions Processing

In-line point source emissions for CMAQ

Processing Pipeline

1. **HMS Fire Detections**
   NOAA satellite-based fire locations
2. **BlueSky Framework**
   Fire behavior & consumption modeling
3. **SMOKE Point Processing**
   Elevpoint for plume rise parameters
4. **Stack Groups**
   Fire location & physical parameters
5. **Inline Emissions**
   Hourly emissions by species

Output Files

Stack Groups:
`stack_groups_ptfire-wild_YYYYMMDD_AIRPACT_04km.ncf`

Inline Emissions:
`inln_mole_ptfire-wild_YYYYMMDD_AIRPACT_04km.ncf`

Key Fire Emissions Species

PM2.5 (fine particles)	NH3 (ammonia)
CO (carbon monoxide)	SO2 (sulfur dioxide)
NOx (nitrogen oxides)	EC/OC (black/organic C)
VOCs (organic compounds)	Acrolein, HCHO

SMOKE Directory Structure

Base: /data/lab/meng/priom_zarrah/SMOKEv5.2/EPM_2022v1/v1/2022hc_cb6_airpact04km/

Input & Intermediate Files

```
inputs/
  └── Inventory, spatial, temporal data

met/2025/AIRPACT_04km/
  └── MCIP meteorology files

scripts/
  ├── onroad/   (RPD processing)
  ├── rwc/      (wood combustion)
  ├── afdust/   (fugitive dust)
  ├── nonroad/  (off-road mobile)
  ├── ptfire/   (wildfire)
  └── merge/    (sector merging)
```

Output Files

```
premerged/
  ├── onroad/RPD/ (31 daily files)
  ├── rwc/        (31 daily files)
  ├── afdust_2025/(31 daily files)
  ├── nonroad/    (31 daily files)
  └── rail/       (31 daily files)

merged_emis/2025/
  └── emis_mole_merged_*
     (Final CMAQ-ready files)

logs/
  └── Processing log files
```

Time Alignment Challenge

Problem

Met files: 08:00-08:00 UTC
(PST-aligned)

Emission files: 00:00-00:00 UTC
(UTC-aligned)

→

Solution

Create PST-aligned emissions:

Hours 8-24 from Day N +
Hours 0-8 from Day N+1

CMAQ Multi-Node Configuration

✓ 3 nodes, 190 processors (19×10 domain decomposition)
✓ Intel MPI with srun and FI_PROVIDER=tcp
✓ 1-day test completed successfully (~94 min/simulated hour)

CMAQ v5.5 with MEGAN Biogenic Emissions

January 2025 Simulation for AIRPACT Domain

Model Configuration

Domain: AIRPACT 4km (285×258 grid)
Period: January 1-13, 2025
Chemistry: CB6r5 with AE7 aerosols
Biogenics: MEGAN v3.2 (inline)
Processors: 190 cores (19×10)
Run time: ~1.5 hours/day

Emission Sources

Anthropogenic:
- EPA 2022v1 Platform
- Onroad, Nonroad, Area sources
- Residential Wood Combustion

Biogenic (MEGAN):
- Isoprene, Monoterpenes
- Soil NO emissions
- Temperature-dependent

Fire Emission:
- BlueSky Framework

Identified Emission Issues

1 Lower Emission

Emissions has lower values at evening/nighttime emissions
RWC peaks during evening hours

2 Semi-volatile POA Treatment

POC → APOC factor in DESID seems lower
POC treated as semi-volatile (evaporates)
Results in lower PM2.5 concentrations

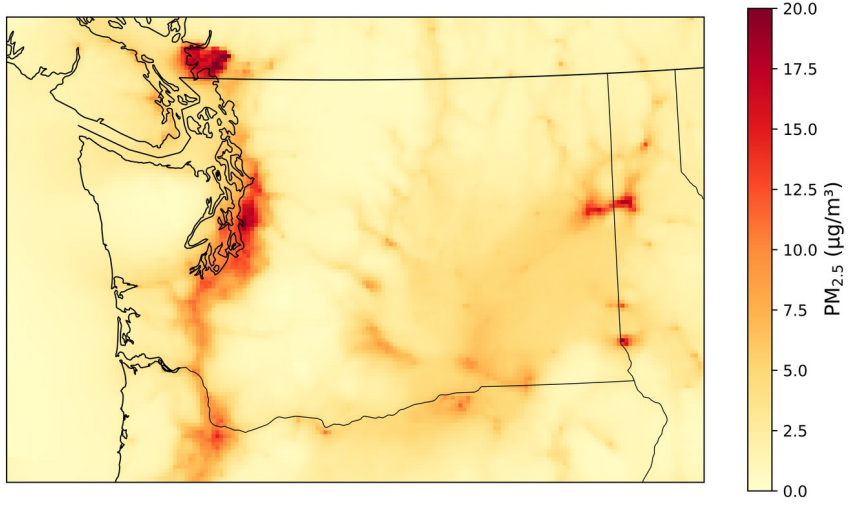


Impact: Model underpredicts PM2.5 by ~22.4% (NMB = -22.4%) at Aberdeen, WA

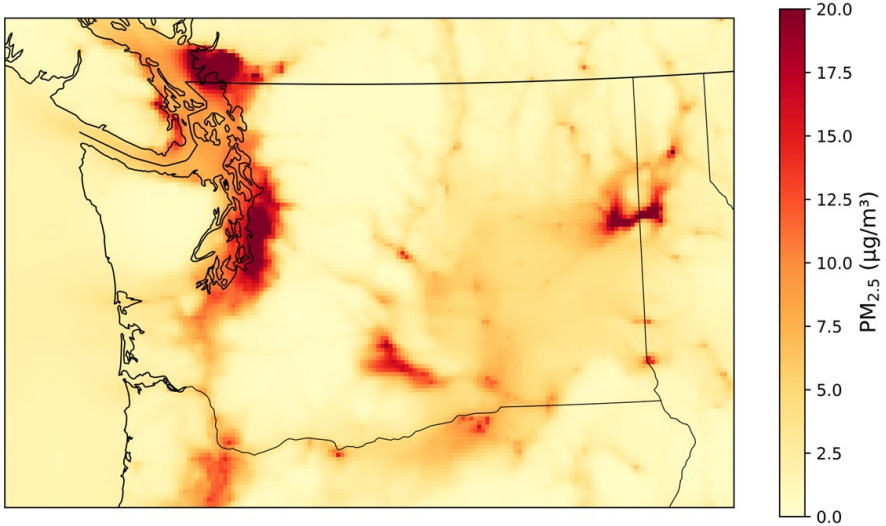
Next Steps: 1) Fix emission file time coverage 2) Verify RWC sector 3) Consider nonvolatile POA option

Jan 2025 10-day Average

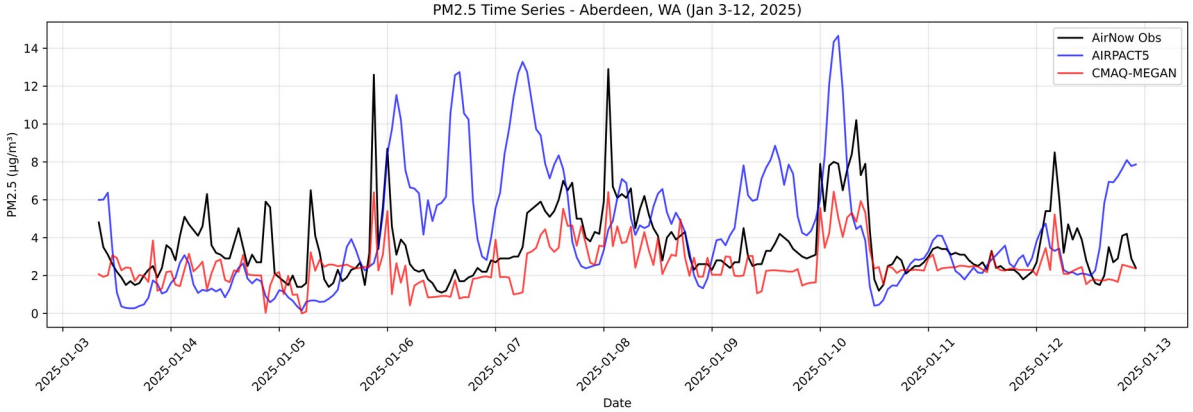
CMAQ w NEI2022

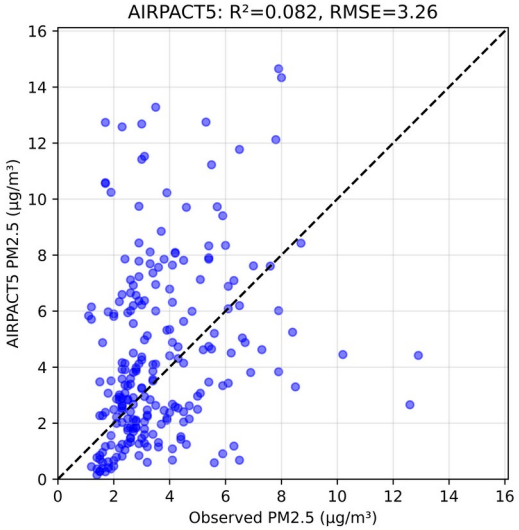


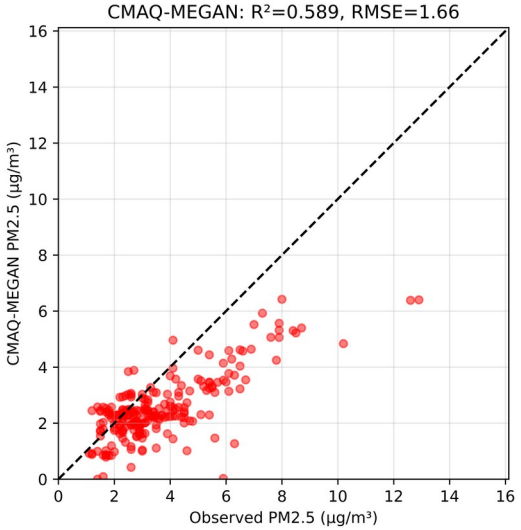
AIRPACT-5



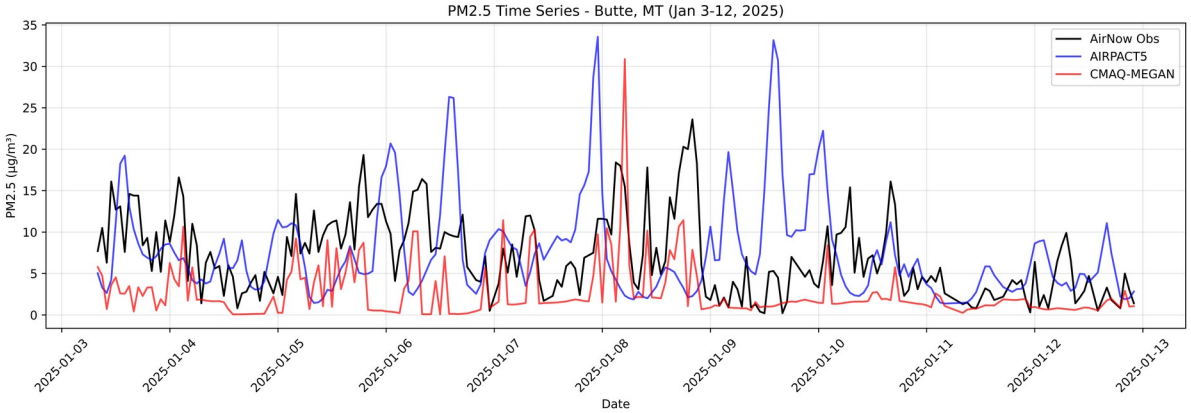
Aberdeen, WA

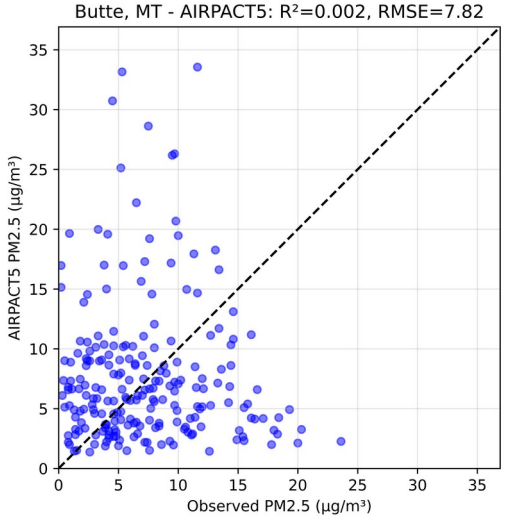

PM2.5 Time Series - Aberdeen, WA (Jan 3-12, 2025)
AirNow Obs
AIRPACT5
CMAQ-MEGAN
PM2.5 (µg/m³)
Date
2025-01-03
2025-01-04
2025-01-05
2025-01-06
2025-01-07
2025-01-08
2025-01-09
2025-01-10
2025-01-11
2025-01-12
2025-01-13

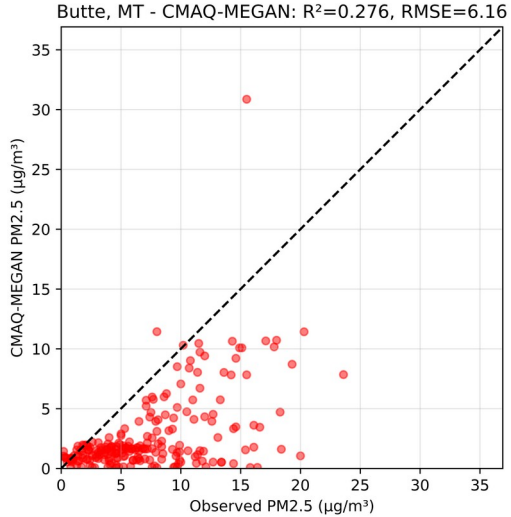

AIRPACT5: R²=0.082, RMSE=3.26
AIRPACT5 PM2.5 (µg/m³)
Observed PM2.5 (µg/m³)


CMAQ-MEGAN: R²=0.589, RMSE=1.66
CMAQ-MEGAN PM2.5 (µg/m³)
Observed PM2.5 (µg/m³)

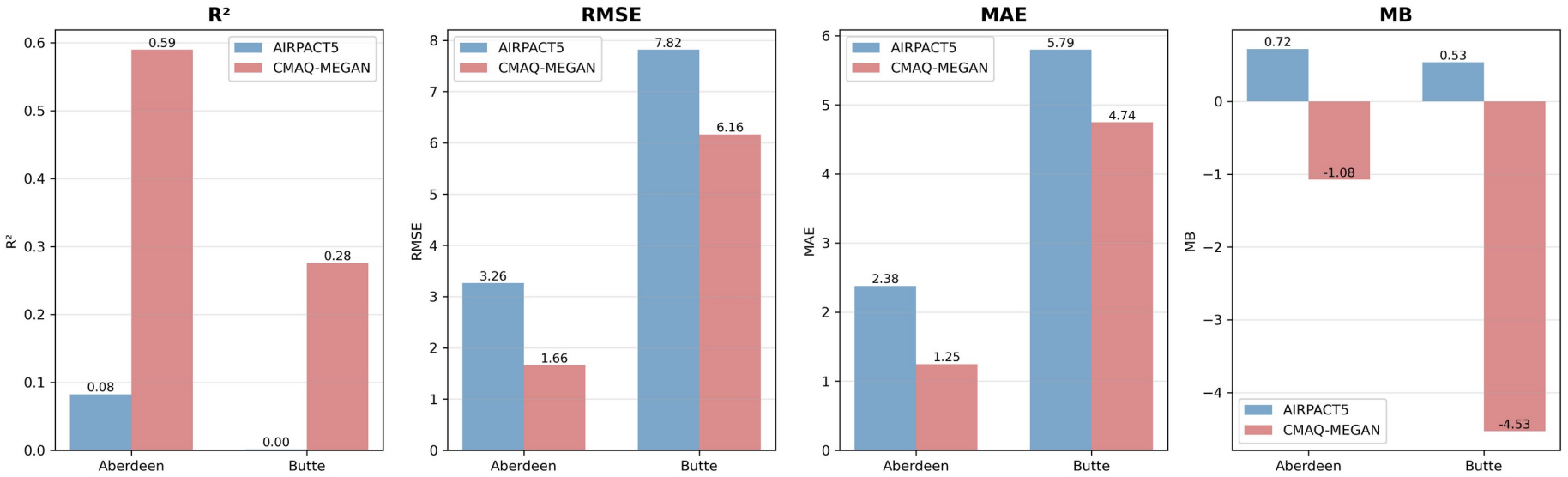
Butte, MT


PM2.5 Time Series - Butte, MT (Jan 3-12, 2025)
AirNow Obs
AIRPACT5
CMAQ-MEGAN
PM2.5 (µg/m³)
Date
2025-01-03
2025-01-04
2025-01-05
2025-01-06
2025-01-07
2025-01-08
2025-01-09
2025-01-10
2025-01-11
2025-01-12
2025-01-13

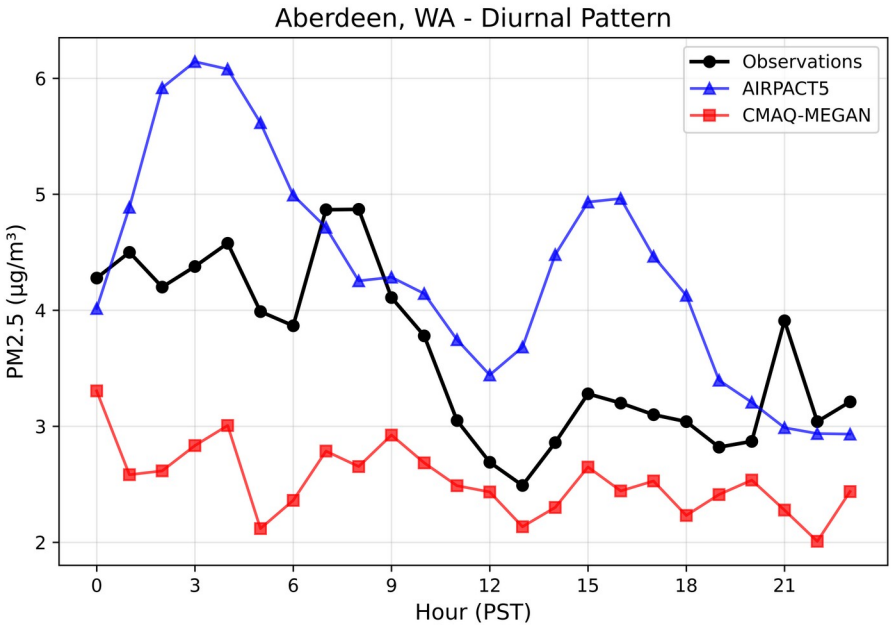

Butte, MT - AIRPACT5: R²=0.002, RMSE=7.82
AIRPACT5 PM2.5 (µg/m³)
Observed PM2.5 (µg/m³)

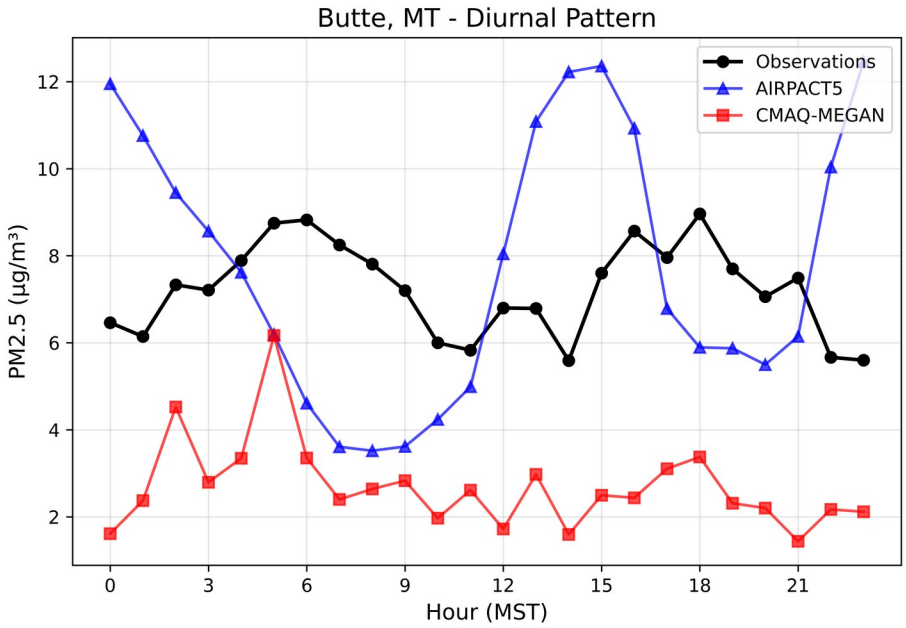

Butte, MT - CMAQ-MEGAN: R²=0.276, RMSE=6.16
CMAQ-MEGAN PM2.5 (µg/m³)
Observed PM2.5 (µg/m³)

Overall Statistics

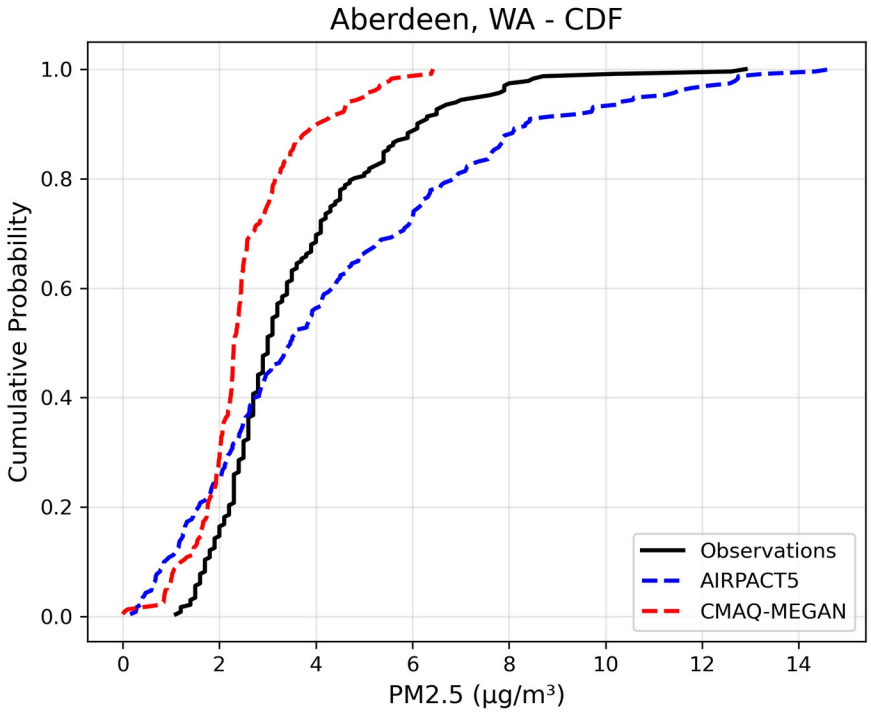


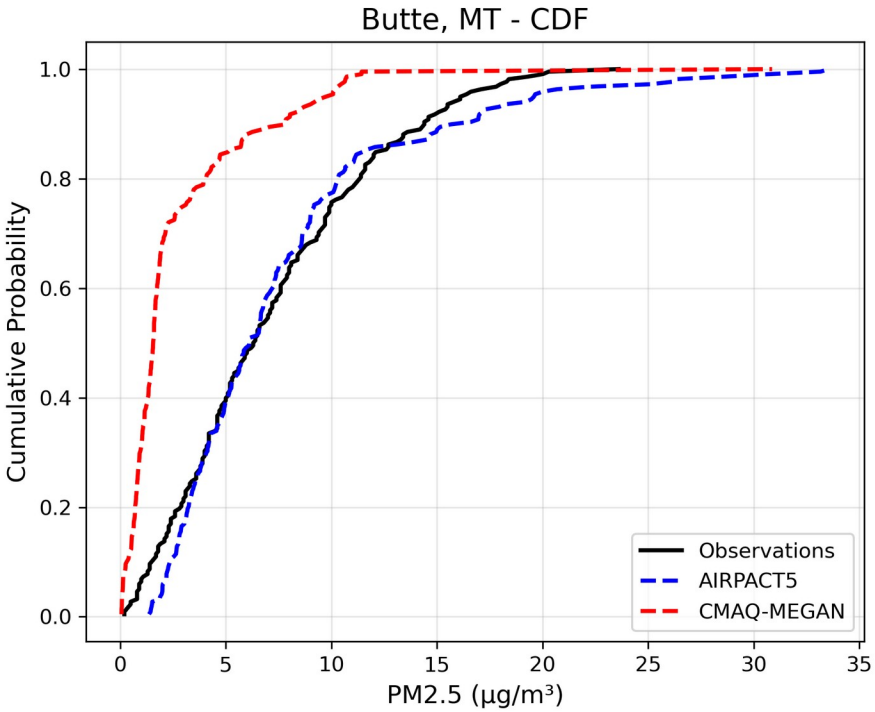
Diurnal Pattern



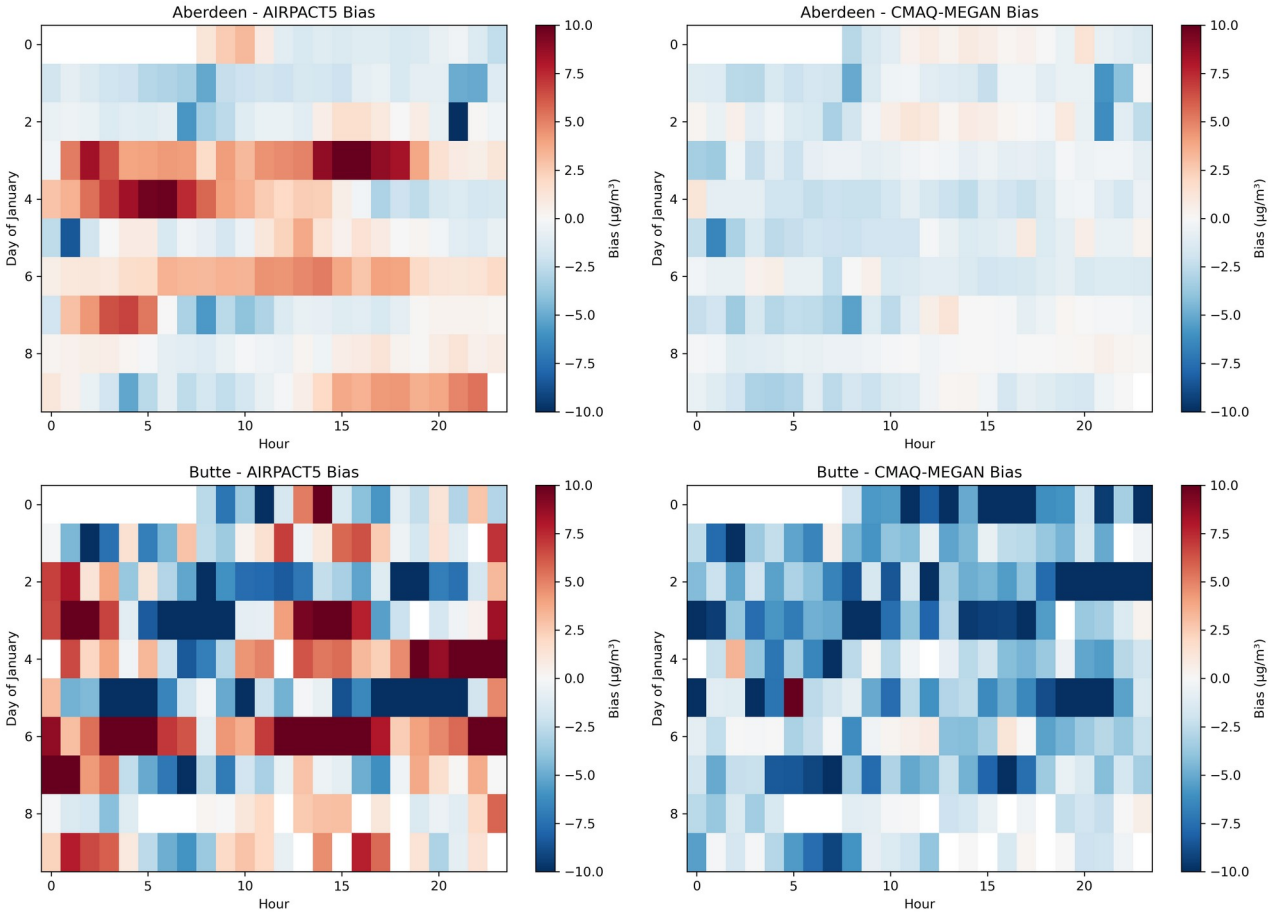


Distribution





Distribution of Bias



Next Steps

1. **Created PST-aligned emissions**

   Python script to combine consecutive days (hours 8-24 + 0-8)

2. **Runing full January and July simulations**

   31 days on Kamiak cluster (~48 hours estimated runtime)

3. **Validate outputs for longer Simulation**

   Compare CMAQ predictions against EPA observations

Next Target: Complete June 2025 CMAQ simulations, and validate.